ONENET · HOSTING SETUP

Provider Console Hosting Setup Guide

Set up your Ubuntu 24.04 + aaPanel server once to host many HR & Payroll clients: wildcard domain and certificate, PostgreSQL, releases, the console, the aaPanel API, scheduled tasks and off-server backups.

1 · DNS + SSL
*.binishaqsoft.com, once

2 · Database
PostgreSQL in aaPanel

3 · Code
Release folder + console

4 · Automate
aaPanel API, Cron, backups

Product
OneNet Provider Console

Server
Ubuntu 24.04 · aaPanel

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What you are building

ADDRESS	SERVED BY	AAPANEL
`admin.binishaqsoft.com`	Provider Console (port 8200)	Node project `provider`, directory /www/hr-provider
`<client>.binishaqsoft.com`	That client's installation (port 9101, 9102, …)	Node project `hr-<client>`, created by the console
`hr.clientcompany.com` (optional)	Same client installation	Extra domain on its project, Let's Encrypt certificate

Everything is done in aaPanel: Node versions in its Node version manager, sites as aaPanel Node projects, domains through Mapping, HTTPS through its SSL, databases on its Databases page and scheduled tasks in its Cron. The existing `attendance.binishaqsoft.com` keeps running as it is.

Requirements and sizing

ITEM	NEEDED
Server	Ubuntu 24.04 with aaPanel (as now). Public IP; ports 80 and 443 open.
Node.js	v20 LTS installed with aaPanel → App Store → Node.js version manager.
PostgreSQL	15 or 16 from aaPanel → App Store (PostgreSQL manager).
Git	`git` (present on Ubuntu; else aaPanel Terminal: `apt install git`).
Memory	About 200–250 MB RAM per client process + 300 MB for PostgreSQL per few clients. 8 GB RAM ≈ 20–25 clients.
Disk	About 1.5 GB per 1 million attendance days, plus 14 backups per client and one release (~600 MB) per version.

> **Large clients**
> A client with 500+ active staff makes manager browsers reload attendance often. Enable gzip for JSON in nginx (aaPanel → Website → site → Config: `gzip_types application/json`) and watch CPU and bandwidth on the dashboard.

Step 1 — Wildcard DNS

At the DNS provider of `binishaqsoft.com` create (once):

TYPE	NAME	VALUE
A	`*` (i.e. *.binishaqsoft.com)	your server IP
A	`admin`	your server IP (optional: the wildcard covers it)

Existing records such as `attendance` and `mcs` keep working: a specific record always wins over the wildcard.

Step 2 — Wildcard certificate

1. aaPanel → Website → add the console site first (Step 6) or any site on the domain, then → **SSL** → **Let's Encrypt**.
2. Choose **DNS verification** with your DNS provider's API (e.g. Cloudflare: API token), and request `*.binishaqsoft.com` and `binishaqsoft.com`.
3. aaPanel stores the files under `/www/server/panel/vhost/cert/<site>/` (`fullchain.pem`, `privkey.pem`) and renews them automatically. Note the exact paths for `WILDCARD_CERT_PATH` and `WILDCARD_KEY_PATH` (Step 6).

Client domains (their own) do not use this certificate; the console requests a Let's Encrypt certificate per domain.

Step 3 — PostgreSQL

1. aaPanel → Databases → **PgSQL** → Add database: name `hr_provider`, user `provider`, a strong password. This is the console's own database (no HR data).
2. Note the PostgreSQL administrator password (PgSQL → root password / postgres). The console uses it to create one role and one database per client: `PG_ADMIN_URL`.
3. Find the PostgreSQL tools folder: aaPanel → Terminal → `ls /www/server/pgsql/bin` (it must contain `pg_dump` and `pg_restore`): `PG_BIN_DIR`.

Step 4 — Folders

```
mkdir -p /www/hr-releases /www/hr-clients /www/hr-backups /www/hr-provider
chown -R www:www /www/hr-releases /www/hr-clients /www/hr-backups /www/hr-provider
chmod 750 /www/hr-clients /www/hr-provider && chmod 700 /www/hr-backups
```

Step 5 — Install the first release

A release is one complete copy of the application. Name it by date, e.g. `2026.11.1`. In aaPanel → Terminal (use the npm of the aaPanel Node version, e.g. `/www/server/nodejs/v20.18.0/bin/npm`, if `npm` is not on the PATH):

```
cd /www/hr-releases
git clone <your repository> 2026.11.1      # or upload a zip with aaPanel → Files and extract
cd 2026.11.1
git checkout <tag or branch>
npm ci
npm run build                 # the HR web app (dist/)
npm run provider:build        # the console web app (provider/dist/)
ln -sfn /www/hr-releases/2026.11.1 /www/hr-releases/current
chown -R www:www /www/hr-releases
```

> **No server address in the build**
> Do not place a `.env.local` with `VITE_API_BASE_URL` in a release folder: hosted builds must use relative addresses so one build serves every client domain.

Step 6 — The console project

Create `/www/hr-provider/.env` from `provider/.env.example` of the release (aaPanel → Files), mode 600:

SETTING	VALUE
PROVIDER_PUBLIC_URL	`https://admin.binishaqsoft.com`
PROVIDER_DATABASE_URL	`postgresql://provider:<password>@127.0.0.1:5432/hr_provider`
PG_ADMIN_URL	`postgresql://postgres:<admin password>@127.0.0.1:5432/postgres`
PG_BIN_DIR	from Step 3, e.g. `/www/server/pgsql/bin`
MFA_ENCRYPTION_KEY	output of `openssl rand -base64 32` (keep a copy off the server)
BASE_DOMAIN / SERVER_IP	`binishaqsoft.com` / your public IP
CLIENTS_ROOT / RELEASES_ROOT / BACKUP_ROOT	`/www/hr-clients`, `/www/hr-releases`, `/www/hr-backups`
WILDCARD_CERT_PATH / WILDCARD_KEY_PATH	from Step 2
AAPANEL_*	Step 8 (can stay empty at first)
SMTP_*	default mail settings copied into each new client

Create `/www/hr-provider/start.cjs`:

```
process.chdir(__dirname);
require('/www/hr-releases/current/provider/server.cjs');
```

1. aaPanel → Website → **Node Project** → Add: name `provider`, directory `/www/hr-provider`, startup file `start.cjs`, port `8200`, Node v20, run user `www`, start with system.
2. Domain (Mapping): `admin.binishaqsoft.com` → Save → Start.
3. SSL: deploy the wildcard certificate (or Let's Encrypt for admin.binishaqsoft.com) and switch on **Force HTTPS**.
4. Recommended: restrict the site to OneNet IP addresses (site → Config / access restriction).

Step 7 — First owner and sign-in

```
cd /www/hr-provider
node /www/hr-releases/current/provider/cli.cjs create-owner you@onenet.pk "Your Name"
```

It prints a temporary password. Open `https://admin.binishaqsoft.com`, sign in, scan the QR code with your authenticator app and confirm the code. Add a second owner under Provider users.

Step 8 — aaPanel API

1. aaPanel → Settings → **API interface** → enable.
2. IP whitelist: add `127.0.0.1` and the server's own IP.
3. Copy the API key into `AAPANEL_API_KEY`; set `AAPANEL_URL=https://127.0.0.1:<aaPanel port>` and `AAPANEL_INSECURE_TLS=true` (aaPanel's own certificate). Set `AAPANEL_NODE_VERSION` to the exact version name shown in the Node version manager.
4. Restart the `provider` project. Server & releases must show **Connected**.

> **Test it once with a throw-away client**
> aaPanel versions differ in their API. Create a test client: if a step (Node project, domain, certificate) does not work through the API, it turns into a checklist with exact values, so setup still completes. Tell OneNet development which step needed manual work.

Step 9 — Template database

Console → Server & releases → **Create template** (or `cd /www/hr-provider && node /www/hr-releases/current/provider/cli.cjs refresh-template`). It takes a few seconds.

Step 10 — Scheduled tasks

aaPanel → **Cron** → Add task → type **Shell script**, run as root or www:

NAME	SCHEDULE	SCRIPT
HR provider nightly	Daily 02:00	`cd /www/hr-provider && /www/server/nodejs/v20.18.0/bin/node /www/hr-releases/current/provider/cli.cjs nightly`

HR provider health	Every 10 minutes	`cd /www/hr-provider && /www/server/nodejs/v20.18.0/bin/node /www/hr-releases/current/provider/cli.cjs health`

Switch on aaPanel's notification for failed tasks.

Step 11 — Off-server backups

1. aaPanel → App Store: install a cloud storage plugin (S3-compatible, Google Drive, FTP…) and connect it.
2. aaPanel → Cron → **Backup directory**: `/www/hr-backups`, daily 03:00, to that storage, keep 14.
3. aaPanel → Cron → **Backup database**: `hr_provider`, daily.
4. Keep an encrypted copy of `/www/hr-provider/.env` and of every `/www/hr-clients/*/.env` off the server (they hold the keys that decrypt backups).

Step 12 — Firewall

aaPanel → Security: allow only 22 (SSH, ideally restricted), 80, 443 and the aaPanel port (restricted to your IP). Never open 8200 or 9101+: the applications listen on 127.0.0.1 and are published by nginx.

Step 13 — First test client

1. Console → New client → name "OneNet Test", address `test1` (any free name), your own email, Lite edition.
2. Follow the job to the end (Mark done for any manual step).
3. Open `https://test1.binishaqsoft.com`, sign in with the temporary password, change it.
4. Create a kiosk setup code and register a kiosk; connect the mobile app with the connection code.
5. Back up now; check the dashboard shows the client as Up. Archive the test client when done.

Install a new release

```
cd /www/hr-releases
git clone <your repository> 2026.12.1 && cd 2026.12.1 && git checkout <tag>
npm ci && npm run build && npm run provider:build
chown -R www:www /www/hr-releases/2026.12.1
```

Then in the console: Refresh template → switch one client → then the others (Admin Guide, "Rolling out a new release"). To upgrade the console itself: `ln -sfn /www/hr-releases/2026.12.1 /www/hr-releases/current` and restart the `provider` project. Old release folders can be deleted once no client uses them.

Kiosk and mobile app builds

- **Kiosk:** build the Windows kiosk from the same release (`npm run kiosk:build`). A kiosk built from this release (or later) registers with setup codes and switches to the fallback address automatically; older kiosks still register with a Kiosk Terminal account. Leave PORTAL_URL empty or set to a default; the setup code decides the server.

- **Mobile app:** build once without `VITE_API_BASE_URL`. On first launch it asks for the company connection code, so one app in the Play Store serves every client.

Troubleshooting

PROBLEM	FIX
Console page: "Provider Console web files are missing"	`npm run provider:build` was not run in the release used by /www/hr-releases/current.
Console does not start: "configuration is missing"	/www/hr-provider/.env missing a value (PROVIDER_DATABASE_URL, PG_ADMIN_URL, MFA_ENCRYPTION_KEY), or start.cjs does not chdir to /www/hr-provider.
Template: "permission denied to create database"	PG_ADMIN_URL must be the PostgreSQL administrator, not the provider user.
New client site shows 502	Its Node project is stopped or crashed: aaPanel → Node Project → hr-<client> → log. Common: Node version mismatch with AAPANEL_NODE_VERSION.
https://<client>.binishaqsoft.com certificate error	The wildcard certificate was not deployed to the project (SSL step) or has not renewed: aaPanel → site → SSL.
Server & releases: aaPanel "IP not in whitelist"	Add 127.0.0.1 and the server IP to the API whitelist.
Release shows "not ready"	`npm ci` or `npm run build` did not finish in that folder.